Case Study 1: Quality Alloys

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Problem Introduction

- **Quality Alloys, Inc. (QA)** is a U.S. distributor of industrial alloys, selling in small, project-based quantities with quick shipping.
- Customers don't maintain an inventory, making single purchases unreliable for forecasting future sales.
- The project analyzed QA's performance from May 2008 to August 2009, using data on financials, sales, and website traffic.
- QA's marketing includes direct mail, magazine ads, and a new web presence to generate leads, but customers can't order online; they must submit manual quote requests.
- The project's objective was to report on the company's performance and recommend improvements.

Analysis of the Problem

- Unique Observations:
  - No information given on what was on QA's marketing and advertisements, so suggestions were open ended
- Problem features:
  - Constraints: limited data and timeframe
  - Complexity: Several different sheets with different information
  - Stakeholders: QA and their marketing/promotion team

Approach

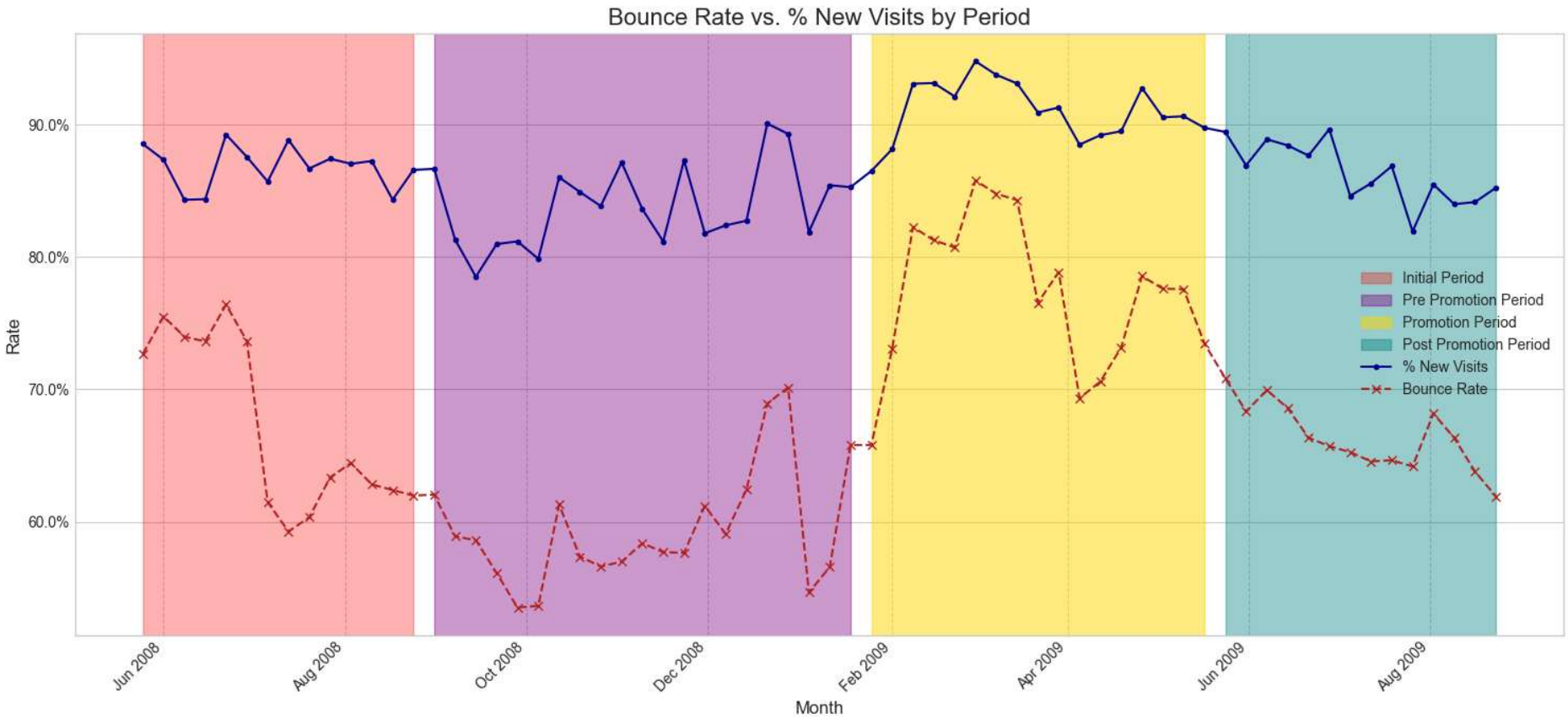
EDA

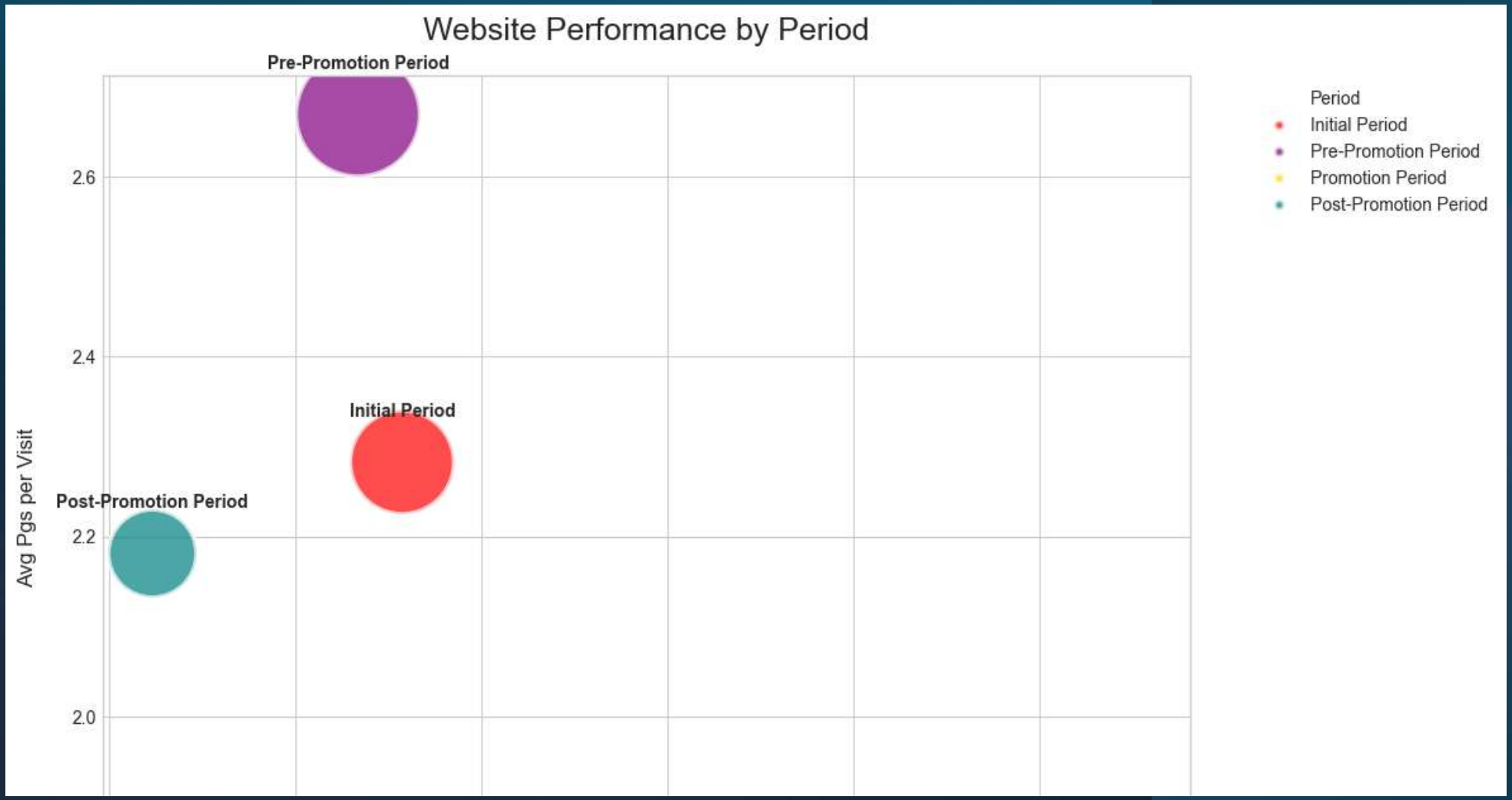
Observing data and deciding what models to create and how to visualize data.

Used Python and libraries like Matplotlib an Seaborn to visualize.

Use the models to draw conclusions and results.

Findings


Bounce Rate vs. % New Visits by Period
Rate
90.0%
80.0%
70.0%
60.0%
Jun 2008
Aug 2008
Oct 2008
Dec 2008
Feb 2009
Apr 2009
Jun 2009
Aug 2009
Month
Initial Period
Pre Promotion Period
Promotion Period
Post Promotion Period
% New Visits
Bounce Rate


Website Performance by Period
Pre-Promotion Period
Initial Period
Post-Promotion Period
Avg Pgs per Visit
2.6
2.4
2.2
2.0
Period
Initial Period
Pre-Promotion Period
Promotion Period
Post-Promotion Period

Verification and Validation

- Verification: Cross check visuals with the given data
- Validation: Conclusions and Recommendations match the objective of the problem and what QA wanted from the analysis

Conclusions

- The website initially had a broad reach, but this was not sustained over time.
- Surprisingly, the highest percentage of website visitors came from South America.
- User engagement was low.
- Site failed to retain visitors.
- The high website traffic did not result in a lasting increase in the pounds of alloys sold.

Recommendations



ENHANCE WEBSITE ENGAGEMENT



ADD AN ONLINE SHOPPING CART



TARGET THE SOUTH AMERICAN MARKET



REPLICATE SUCCESSFUL MARKETING